Observing and Responding to DOM Changes

If we want to use the DOM as a place to hold on to the state of the client-side system, it's helpful to be able to know when that state changes. In a Hotwire system, where new HTML text might be added to the DOM as a result of a form submission or an ActionCable text receipt, being able to detect that HTML change and perform an action based on it allows for a lot of cool behavior, even while keeping the drawing logic on the server side.

We can do this detection by using the MutationObserver API, which is a standard part of every recent browser. Stimulus itself is powered by MutationObserver calls, so any browser that can run Stimulus can manage MutationObservers. Browsers that are older than, say, Internet Explorer 11, need the @stimulus/polyfill package to run.

For our purposes here, we're going to use a MutationObserver object to solve a minor problem in our schedule page. When we list favorite concerts, it'd be nice if that list were sorted by the date of the concert. The problem is that even if we sort the initial server-side list, as new favorites come in via Turbo Stream, we can only append them. What we want is to have them go into the list in their correctly sorted place. To do this, we're going to use the MutationObserver API to build a generic Stimulus controller that will sort its internal targets whenever a new target is added. (This code is a somewhat simplified version of what Hey.com uses to sort emails when they are added back into the "already seen" list.)

This is the logic here:

1. Detect when a new item has been added to the list, and then
2. Ensure the item list is properly sorted.

To make this work, we need to make a couple of minor changes in the ERB file for the favorite list. Specifically, we need to make some changes to the actual list of favorites: